Kaalchakra — Progress Report

Every period, in its place. · clash-free semester timetables for any college

Punjab Jobs AI Bootcamp · Hackathon 1 · as of 21 Oct 2026 · github.com/Navkaur-codes/hackathon-1 · React + Vite + TypeScript, client-side

v0.3	184	0	96 / 89	0.4 s	1,617 h
current version (v0.1, v0.2 done)	automated tests all passing	clashes in any timetable (validator-checked)	quality score DIET / DCAS demos	20 candidate timetables solved in the browser	contact hours per semester (DIET, 105 days)

What works today

Area	Built and tested
Multi-college data	Any shape of college from configuration — departments, years, sections × strength, subjects per programme, rooms (own / shared), teachers; nothing hard-coded. Two demo colleges of different shapes (engineering Mon–Sat × 7 slots; arts & science Mon–Fri × 6 slots).
Breaks & semester	Staggered lunch per department / year / section / day; shorter lunch slots. Real semesters (dates, holidays, exam days): the weekly pattern is rolled out over every teaching date with contact hours per subject.
Scheduling engine	Dynamic most-constrained-first solver; 14 hard rules never broken (teacher / section / room clashes across departments, room type & size & ownership, availability, breaks, lab blocks, daily/weekly limits, teacher lunch); 20 seeded tries keep the best; runs in a Web Worker.
Trust & explanations	Independent validator ("0 clashes — verified"); 0–100 quality score from 7 soft goals; pre-flight check names impossible input; unplaced classes come with a plain reason and a fix; repair step as safety net.
Leave cover (engine)	Leave is for one date: the best free teacher is auto-allotted per class (same dept → same subject → knows the section → lightest load), never breaking limits or lunch; weekly pattern unchanged.
Screens	Landing with live demo cards → Generate → result summary; timetable page with Section / Teacher / Room grids (real times, hatched breaks, striped labs, subject colours, phone-friendly); "Why here?" checklist for every class (built, awaiting push).

Roadmap

Version	Scope	Status
v0.1 Foundation	Theme, data model, structure generator, demo colleges	Done
v0.2 Engine	Validator, solver, semester, leave cover, pre-flight, score, worker	Done
v0.3 Timetable views	Grids (done) · Clash Court "why here?" (built, awaiting push) · motion + print (next)	In progress
v0.35 Finder & filters	Who's free (teachers / rooms), common free slot, filter bar, Ctrl+K	Planned
v0.4 Super & College Admin	Saved data, role + college switcher, colleges list, Structure Builder, setup + paste-from-Excel, publish	Planned
v0.5 HOD + Room Exchange	Department data, Draft → Ready, department-by-department generate, lend / borrow rooms	Planned
v0.6 Teacher & Student	My Week / Today, leave request, real-date view with substitutes, Now / Next, Meet-a-Teacher	Planned
v0.7 Ops & insights	Leave inbox & substitute review screens, lock & regenerate, Conflict Center, dashboard, shared-room ledger	Planned
v0.8 Kaal Assistant	Rule-based chatbot for small questions (free teachers, rooms, next class)	Planned
v0.9 Authentication	Login, admin-managed users, hashed passwords, page guards	Planned
v1.0 Ship	Full test run, live checklist, README, submit	Planned

Open items

• **Vercel** not connected yet — no live link. • Timetables are not **saved** until v0.4 (refresh returns to the landing). • Teacher daily **balance** is the weakest quality part (75% / 65%). • Scope added on request so far: subjects, staggered breaks, semester dates, date-specific leave with auto substitutes (pulled forward from v0.7). • If time runs short, drop in order: share link → Ctrl+K → ledger → lock & regenerate → dark mode → auth.